GG-S-935A
June 16, 1977
SUPERSEDING
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June 25, 1959

FEDERAL SPECIFICATION

SYRINGES, HYPODERMIC, NEEDLE LOCK, GLASS, 3 ml (cc)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers Luer lock type syringes suitable for use in medical procedures.

1.2 Classification.

1.2.1 Type and size. Syringes shall be of two types and one size as specified (see 6.2).

Type I - Syringes without finger rings, 3 ml (cc).
Type II - Syringes with finger rings (control), 3 ml (cc).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposals form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Set-up.

FSC 6515

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Federal Standards:

Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547	-	Liners, Case and Sheet, Overwrap; Waterproof, Flexible.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

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American National Standards Institute (ANSI) Standard:

Z70.1 (- American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute Incorporated, 1430 Broadway, New York, NY 10018.)

United States Pharmacopeial Convention, Inc. (USP):

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

American Chemical Society Publications:

Reagent Chemicals.

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth Street, N.W., Washington 6, DC).

American Iron and Steel Institute (AISI) Publication:

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street, N.W. Washington, DC 20036.)

American Society for Testing and Materials (ASTM) Standard:

C-693 - Density of Glass by Buoyancy.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Barrel and plunger. Barrel and plunger shall be fabricated of high-grade, heat-resistant, corrosion-resistant borosilicate glass throughout.

3.1.1.1 Density. The glass shall have an average density of not more than 2.4 grams/cc when tested according to 4.4.10.

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3.1.1.2 Alkalinity. Syringes shall be tested for alkalinity as specified in 4.4.3. The volume of 0.02 Normal (N) sulfuric acid required to neutralize the pink color, which may be observed after the solution has cooled, shall not exceed 0.03 ml per syringe.

3.1.1.3 Annealing. Syringes shall be examined for strain as specified in 4.4.8. A moderate amount of evenly distributed longitudinal strain is acceptable. Any ring strain or severe longitudinal strain shall not be acceptable.

3.1.1.4 Permanency of marking. Syringes shall be subjected to the permanency of marking test as specified in 4.4.9. Syringes shall be considered to pass the test if not more than one-half of any graduation line or numeral has been lost, and the area showing complete pigment loss is less than 2 percent of the total pigmented area, and if the legibility of any portion of the marking is not seriously impaired.

3.1.2 Luer-lock tip. The Luer-lock tip shall be brass with nickel, or chromium over nickel plating.

3.1.3 Side and top rings, locknuts, side and top-ring extensions, and top collar. The side and top rings, locknuts, side and top-ring extensions, and top collar shall be brass, chromium plated over nickel.

3.1.4 Washer. Washer shall be fabricated of block tin.

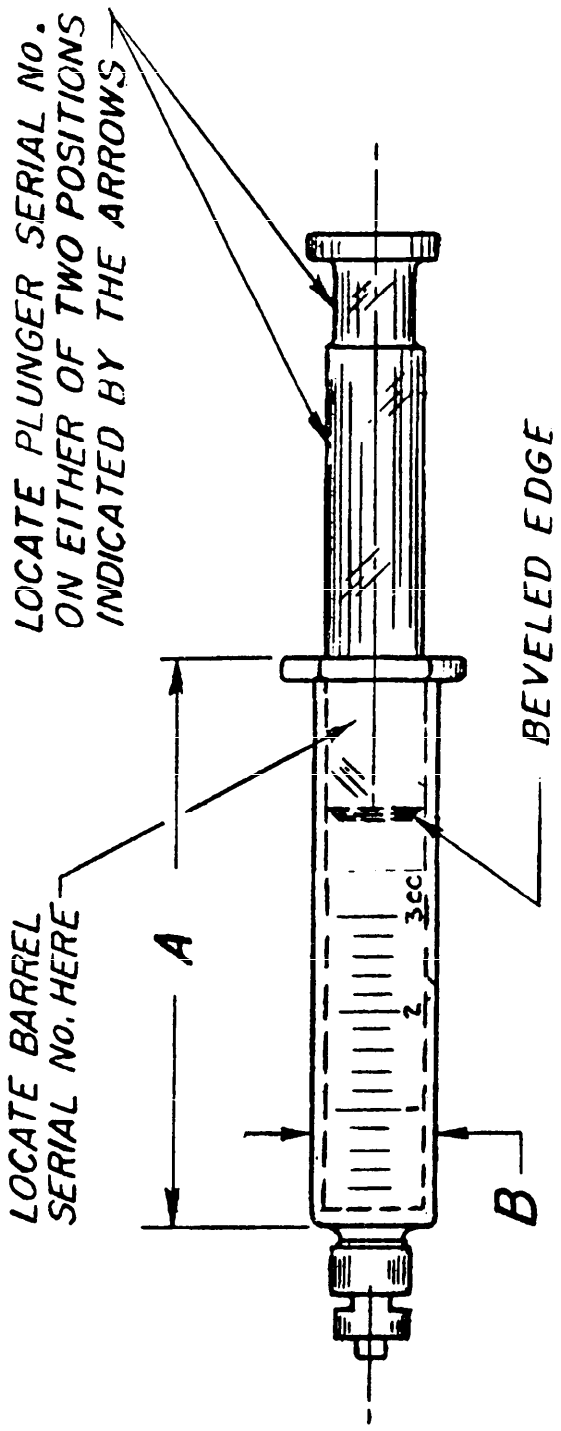
3.1.5 Plunger retainer spring. Plunger retaining spring shall be fabricated of American Iron and Steel Institute (AISI) 304 stainless steel.

3.2 Construction.

3.2.1 Type I. Syringe shall consist of a barrel, plunger, and Luer-lock tip. Except as hereinafter specified, syringe construction shall be as shown in figure 1. The Luer-lock tip shall be constructed as shown in American National Standards Institute (ANSI) Standard Z70.1.

3.2.1.1 Barrel. Barrel shall be a tube closed at one end except for an orifice, and open at the other end. Barrel shall have attached thereon at the orifice end a Luer-lock tip for attaching a syringe needle. The wall of the open end shall be thickened and extended radially outward to form a flange to prevent slipping of the barrel through the fingers. The inside surface of the barrel shall be finished to a very fine uniform ground surface,

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SIZE	DIMENSIONS OF BARREL		
	A	B	WALL THK. MIN.
1(3cc)	$3\frac{13}{64} + \frac{7}{64} - \frac{}{}$	$\frac{13}{32} + \frac{3}{32} - \frac{}{}$	.035

FIGURE 1.—Syringe without finger rings.

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or a smooth, clear unground surface, free from pits, grind marks, or open-air lines. There shall be no high spots along the periphery where the bottom joins the interior wall of the barrel. The distal end of the plunger, when fully inserted, shall mate evenly with the distal end of the barrel.

3.2.1.2 Flange. The form and dimensions of the flange shall be appropriate to the syringe and shall extend outward from the wall, sufficient to give ample fingerhold, and lengthwise of the barrel sufficient for strength and durability. Opposite sides of the flange shall be flat to provide straight parallel sides of ample length to prevent rolling.

3.2.1.3 Plunger. Plunger shall be hollow. The distal end of the plunger shall be ground flat, or formed (unground) flat, and at right angles to the axis, and its edge shall be slightly beveled. A permanent mark of dark color shall be present on the beveled edge, or the plunger may be of dark colored glass throughout its length, or for part of its length from the distal end. The distal end of the plunger shall be clearly visible and accurately indicated through the barrel wall. The proximal end of the plunger shall extend well out of the barrel and be shaped to form a button-type knob. The sides of the plunger shall show, under low power (7X) magnification, a uniform, smooth-ground surface, free from pits, grind marks, or open-air lines, and shall accurately fit its individual barrel so as to make the syringe successfully withstand the applicable leak tests, specified in section 4, and yet move freely throughout its excursion in the barrel. Pigment shall correspond to that used for barrel markings.

3.2.1.4 Syringe dimensions. Dimensions of syringe, not including the Luer-lock tip, shall be in conformance with figure 1. Dimensions of the Luer-lock tip shall be in conformance with ANSI Z70.1.

3.2.1.4.1 Luer-lock tip. The Luer-lock tip shall make a secure, watertight union with a needle hub or needle adapter having a corresponding taper and shall be of such dimension and taper as to meet the requirements specified in 3.4.2, and 3.4.3. The tip shall be joined to the barrel in such a manner as to form a strong union with the barrel. The orifice through the tip shall have a minimum diameter of 0.070 inch, shall be concentric therewith and shall be as shown in ANSI Z70.1. No adhesive shall be used to hold the metal tip to the glass tip. There shall be no air space between the glass tip and metal tip. Strength of joint between tip and glass shall be adequate to meet the test specified in 4.4.4. When tested as specified in 4.4.1, tapered

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portion of the tip shall fit the ring gage within the "GO" - "NO GO" limits and there shall be no movement of the ring gage. In addition, when the tip is tested with the thread checking gage as specified in 4.4.1, the gage shall be capable of entering the tip and being fully seated.

3.2.2 Type II. Syringe shall consist of a barrel, plunger, side ring assembly, top-ring assembly, and Luer-lock tip. Syringe shall be constructed as shown in figure 2. The Luer-lock tip shall be constructed as shown in ANSI Z70.1.

3.2.2.1. Barrel. Barrel shall be as specified in 3.2.1.1, except that the proximal end shall have a beaded edge instead of a flange.

3.2.2.2 Plunger. Plunger shall be as specified in 3.2.1.3 except that the proximal end shall not be shaped to form a button-type knob. Instead a top ring extension (with a top ring) shall be securely attached to the proximal end.

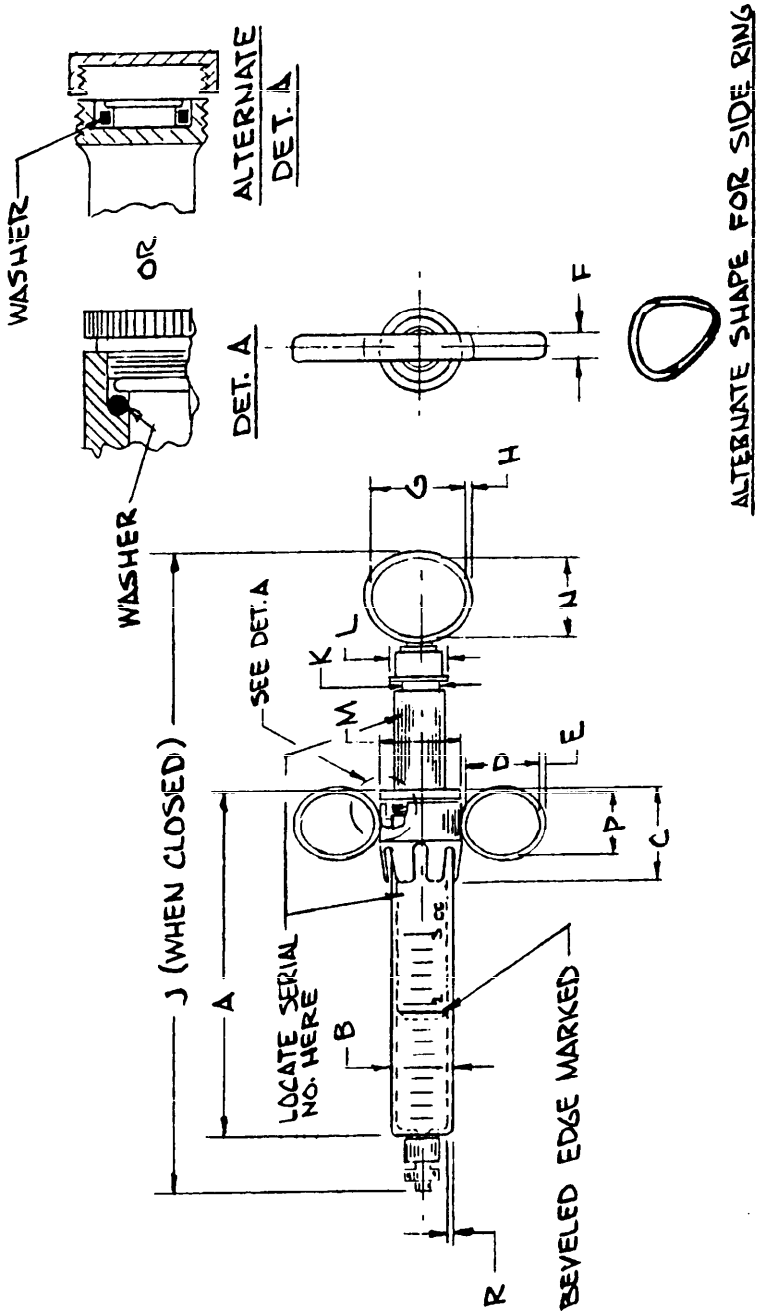
3.2.2.3 Finger ring assemblies. Finger ring assemblies shall be of the following construction:

3.2.2.3.1 Side rings. Two rings shall be brazed to a side ring extension on diametrically opposite sides of the extension. The extension shall be equipped with a block tin or rubber washer and shall be formed with barrel centering slots on its distal end as shown in figure 2. The top collar, holding the side-ring extension in position at the proximal end of the barrel, shall be threaded to fit the threaded portion of the inside of the proximal end of the side ring extension, and shall be undercut to retain a pronged spring which shall act as a plunger brake.

3.2.2.3.2 Top ring. A ring, as shown in figure 2, shall be brazed to one end of threaded top-ring extension. This ring extension shall have an upper and lower locknut threaded on it and shall be cemented or locked by means of a flange nut onto the proximal end of the plunger. Top ring extension shall not be loosened by sterilization or boiling. The locknuts shall be so positioned on top-ring extension that when the syringe is assembled, with plunger pushed to the bottom of the barrel, the extreme distal end of the plunger shall register precisely with the zero graduation of the barrel scale.

3.2.2.4 Syringe dimensions. Dimensions of syringe, not including the Luer-lock tip, shall be in accordance with figure 2. Dimensions of the Luer-lock tip shall be in accordance with ANSI Z70.1.

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SIZE	A	B	C	D (MIN.)	E	F	G (MIN.)	H	J (MAX.)	K O.D.	L	M	N (MIN.)	P (MIN.)	R (MIN.)
1(3CC)	$3\frac{3}{16} \pm \frac{1}{16}$	$\frac{29}{64} \pm \frac{1}{16}$	$1 \pm \frac{1}{16}$	$\frac{27}{32}$	$\frac{1}{16}$	$\frac{9}{32}$ TO $\frac{5}{16}$	$\frac{15}{16}$	$\frac{1}{16}$	$5\frac{13}{16}$	$\frac{9}{32}$ TO $\frac{3}{8}$	$\frac{9}{16}$ TO $\frac{3}{4}$	$\frac{21}{32}$ TO $\frac{15}{16}$	$\frac{15}{16}$	$\frac{27}{32}$	.035

UNLESS OTHERWISE SPECIFIED TOLERANCES SHALL BE ± 1/64.

NOTE: DO NOT SCALE FIGURE.

FIGURE 2. -- Syringe with finger rings

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3.2.2.4.1 Luer-lock tip. The Luer-lock tip shall be as specified in 3.2.1.4.1.

3.3 Graduations and other markings. The outside surface of the barrel of types I and II syringes shall be marked with the graduation scale, the capacity of the syringe, serial number (corresponding to serial number on plunger), the name or registered trademark of manufacturer or contractor. All of these markings shall be applied to the barrel in such a manner that they are in line with flattened side of the flange. Main graduation marks shall be at least 33-1/3 percent longer than the subgraduation marks, and the "0" mark of the scale shall register precisely with the extreme distal end of the plunger or with the dark colored mark on the bevel, when the plunger is pushed to the bottom of the barrel. All markings shall be clearly legible and either fused in, fused on, or etched and fused.

3.3.1 Capacity. The syringe shall be of 3 milliliters (cubic centimeters) capacity with main graduations and numbers at each ml (cc), except that no number is required at zero. There shall be subgraduations at each 1/5 ml (cc) interval.

3.4 Fluid tightness.

3.4.1 Fluid tightness between plunger and barrel. The syringe shall be tested for fluid tightness between the plunger and barrel as specified in 4.4.2.1. During the first test, there shall not be enough water to pass the plunger to produce a falling drop. After each successive test, any water that may have passed the plunger shall be wiped away and the next test applied, during which there shall not be enough water to pass the plunger to produce a falling drop.

3.4.2 Fluid tightness Luer-lock tip to hypodermic needle hub (or needle adapter). The Luer-lock tip shall be tested for fluid tightness between the tip and a hypodermic needle hub, as specified in 4.4.2.2. When pressure is applied for 10 seconds, there shall be no leakage exceeding one falling drop between the syringe tip and the hub of the gage.

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3.4.3 Fluid tightness and freedom from air space between Luer-lock tip and glass tip. The Luer-lock tip shall be tested for fluid tightness and freedom from air space between the Luer-lock tip and the glass tip as specified in 4.4.2.3. When the pressure is applied for 20 seconds, there shall be no leakage between the Luer-lock tip and the glass tip. In addition, the syringe barrel shall be examined by viewing through the inside of barrel from the flanged end and there shall be no evidence of air space between the Luer-lock tip and the glass tip (i.e., presence of colored solution).

3.5 Security of attachment of metal tip to syringe. When tested as specified in 4.4.4, the tip shall not become detached nor show signs of loosening.

3.6 Quality of plating. When given the boiling water exposure as specified in 4.4.5, there shall be no blistering or flaking of plating and no corrosion of the basis metal.

3.7 Resistance to autoclaving. The syringes shall be tested for resistance to autoclaving as specified in 4.4.6. At the time the test is completed (while syringes are still hot), the plunger shall be freely movable in the barrel.

3.8 Resistance to thermal shock. The syringes shall not break (due to temperature changes) when submitted to the thermal shock test, as specified in 4.4.7.

3.9 Accuracy. When tested as specified in 4.4.11, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.10 Serial numbers. Plunger and barrel shall be permanently marked near their proximal end with a corresponding serial number. There shall be no duplication of serial numbers among syringes of the same type and size delivered on any particular contract.

3.11 Instruction sheet. Each unit shall be supplied with an instruction sheet containing a statement that solutions used for cleaning shall be "neutral" having a pH of 6 to 8.

3.12 Workmanship. Workmanship shall be first class throughout. Luer lock syringes shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the glass syringe, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in tables I and III. Unit of product for examination purposes shall be one syringe.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (Percent defective)
Major	I	1.0
Total	I	5.0

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table II. Unit of product for test purposes shall be one syringe. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Component	Characteristic	Requirement	Test procedure	Insp. level	AQL (percent defective)
Glass	Density	3.1.1.1	4.4.10	S-1	1.5
	Alkalinity	3.1.1.2	4.4.3	S-1	2.5
	Annealing	3.1.1.3	4.4.8	S-1	2.5
	Permanency of marking	3.1.1.4	4.4.9	S-1	2.5
Barrel	Luer-lock tip	3.2.1 3.2.1.4 & 3.2.2	4.4.1	S-1	1.5
	Security of attachment	3.5	4.4.4	S-2	1.5
Syringe	Dimensions	3.2.1.4 & 3.2.2.4	FIGS 1 & 2	S-1	1.5
	Fluid tightness	3.4	4.4.2	S-2	1.5
	Plating	3.6	4.4.5	S-1	1.5
	Resistance to autoclaving	3.7	4.4.6	S-1	2.5
	Resistance to thermal shock	3.8	4.4.7	S-1	2.5
	Accuracy	3.9	4.4.11	S-1	1.5

4.3 Examination.

4.3.1 Examination of syringes. The syringes shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any syringes in the sample containing one or more defects shall be rejected, and if the number of defective syringes in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Major:	
101	Metal components missing.
102	Plunger or barrel missing.
103	Graduation markings incomplete, incorrect, or illegible.
104	Plunger not dark colored at distal bevel.

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TABLE III. Classification of defects (cont'd)

Categories	Defects
105	Plunger stuck.
106	Plunger does not operate smoothly in barrel.
107	Plunger or barrel cracked or chipped.
Minor:	
201	Name or registered trademark of manufacturer missing.
202	Instruction sheet missing.

4.3.2 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE IV. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Container	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of container.

4.3.2.1 Examination of closure, waterproofing and banding of container. When the shipping container is required to comply with PPP-B-636, PPP-B-585, PPP-B-601 or PPP-B-621 the examination for defects in closure, waterproofing or banding shall be in accordance with the appendix of the box specification.

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4.4 Tests. Tests shall be conducted as necessary, to determine, compliance with specification requirements.

4.4.1 Taper of Luer-lock tip. The Luer-lock tip shall be tested with the ring gage as shown in ANSI Z70.1 This gage shall be slipped onto the tip with light pressure. In addition when this ring gage is fitted onto the tip as specified above, any movement of the gage on the tip, when the gage is pressed lightly from side to side in all directions, shall be noted. The Luer-lock tip shall also be tested with the thread checking gage shown in ANSI Z70.1. (This gage shall be as shown in ANSI Z70.1 and shall be constructed with a suitable fluid path, throughout its length, to permit performance of the fluid tightness tests specified in 4.4.2.2 and 4.4.2.3). The thread checking gage shall be fitted to the tip with light pressure.

4.4.2 Fluid tightness test. The syringe shall be attached to a testing machine having a water pressure gage. The water pressure in pounds per square inch (psig) applied to the syringe shall be 45 psig.

4.4.2.1 Fluid tightness between plunger and barrel. Pressure used shall be as specified in 4.4.2. The water used as a testing fluid shall be let in through the tip of the syringe. The plunger shall be drawn to a point one-fourth of the distance of the graduations where it shall be allowed to remain for 10 seconds. The plunger shall then be drawn back to midway between first position and the top graduation mark where it shall be allowed to remain for a similar length of time. The plunger shall then be drawn back to the top graduation mark where it shall be allowed to remain for a similar length of time. The syringe shall be tested in a horizontal position.

4.4.2.2 Fluid tightness Luer-lock tip to hypodermic needle hub. A thread checking gage as shown in ANSI Z70.1 (and having a suitable fluid path throughout its length) shall be firmly fitted to, and fully seated (locked) with the Luer-lock tip of the syringe. The syringe shall be attached to the water pressure source described above in 4.4.2. The water shall be let in until the pressure is raised to the value specified in 4.4.2 with the plunger held stationary at the halfway point. The syringe shall be allowed to remain under this pressure for 10 seconds.

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4.4.2.3 Fluid tightness and freedom from air space between the Luer-lock tip and the glass tip. The syringes (dry) shall first be subjected to autoclave sterilization for 15 minutes at 15 psig steam pressure. After cooling, the syringe shall be filled to a minimum of one-fourth of the syringe capacity with an aqueous solution of red or green vegetable dye or a 1 percent solution of sodium fluorescein (adjusted with sodium hydroxide to a pH of approximately 9). The thus filled syringe shall be attached to a pressure source capable of producing a pressure of 45 psig inside the syringe. The pressure shall be raised to 45 psig and the syringe shall be allowed to remain under this pressure for 20 seconds. At the end of this time the colored fluid shall be let out of the barrel, the plunger shall be removed from the barrel, and the syringe barrel shall then be rinsed with tap water to remove the colored solution from the barrel surfaces. The examination for entrapped fluid shall be performed by filling the barrel to approximately one quarter of the syringe capacity with an 80 percent aqueous solution of sucrose (cane sugar) and viewing through the inside of the barrel from the flanged end. In the event of doubt or disputes as to the presence of colored solution, the Luer lock tip shall be removed for a more positive determination.

4.4.3 Alkalinity. Syringes shall be washed in distilled water. All surfaces shall be brushed to remove loose particles and thoroughly rinsed with the water. Unassembled syringes, with metal components removed, shall then be placed in an alkali-free resistant glass beaker and covered with a solution of freshly boiled distilled water to which has been added 0.2 cc. of United States Pharmacopeia (USP) phenolphthalein test solution for each 100 cc. of water. Five syringes shall be used with a 400 cc beaker containing 225 ml of water. Syringes shall be autoclaved at 15 psig steam pressure for 15 minutes and then removed from the autoclave. After the solution has cooled, any pink color which may be observed shall be neutralized (titrated) with 0.02N sulfuric acid.

4.4.4 Security of attachment of metal tip to syringe. The barrel of the syringe shall be firmly held in a suitable fixture with the tip free and exposed. A tensile load of 25 pounds shall be applied to the metal tip and allowed to remain for 5 minutes.

4.4.5 Boil test. The plunger shall be removed from the barrel and the plunger and barrel with all metal components present on them shall be immersed in distilled water contained in a borosilicate glass vessel. The water shall be heated to boiling and the boiling maintained for a half-hour after which the heat input shall be stopped. After an additional 24 hours, the syringe shall be removed from the water and examined.

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4.4.6 Resistance to autoclaving. Syringes (dry), with the plunger pushed completely in the barrel, shall be subjected to autoclave sterilization for 15 minutes at 15 psig steam pressure.

4.4.7 Resistance to thermal shock. Syringes shall be transferred suddenly from boiling water into ice water (3° to 5° centigrade). This shock test shall be repeated at least 10 times.

4.4.8 Annealing. Syringes shall be examined for strain with a polariscope to determine compliance with the requirements of 3.1.1.3.

4.4.9 Permanency of marking test. All metal parts shall be removed from the syringes prior to test. The syringes shall be completely immersed in an open beaker containing hydrochloric acid solution, as described below. Autoclave the beaker containing the hydrochloric acid and syringes at 15 psig and 250° Fahrenheit (F) for two periods of 30 minutes each. The 30-minute period shall begin at the instant the temperature of the hydrochloric acid solution in the autoclave reaches 245°F in the liquid. At the end of the 30-minute period, remove the syringes, rinse with cold tap water, and dry by wiping with cellulose tissue. Repeat the autoclaving operation and drying process, and then examine for loss of pigmentation. The hydrochloric acid solution shall be as follows:

Hydrochloric acid, 35-38 percent, USP or American Chemical Society (ACS) quality - 10.0 milliliters (ml.)

Distilled water, quantity sufficient to make 1000 ml.

4.4.10 Density. Density of glass shall be determined by the method of ASTM Standard C-693.

4.4.11 Accuracy test. Syringes shall be tested (with water) for accuracy at each main graduation mark (see 3.3.1, 3.3.2, and 3.3.3) with a calibrated burette.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Unit package. Each Type I or Type II syringe along with instruction sheet (see 3.11) shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.1.2 Intermediate package. Twelve packaged type I or type II syringes shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of the applicable box specification.

5.1.2 Level C. Syringes with instruction sheets (see 3.11) packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Two hundred eighty eight packaged syringes, type I or type II as specified, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendis thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Two hundred eighty eight packaged syringes, type I or type II as specified, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged syringes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be as specified in Fed. Std. No. 123.

5.3.2 Military activities. Each unit, intermediate package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. Syringes covered by this specification are intended for use in medical procedures to inject drugs in solution and other liquids through a hollow needle, of samll bore into subcutaneous tissue; the metallic tip and locking device are incorporated to hold the needle firmly in place. The control syringe (Type II) incorporates thumb and finger rings attached to the plunger and barrel of the syringe to facilitate ease of use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Applicable type (see 1.2).
(c) Applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Marking required (see 5.3).
(e) Applicable NSN.

6.3 NSN coverage. This specification covers the following items(s) appearing in the Federal Supply Catalog:

National Stock No.	Item Identification	Specification Designation
6515-00-380-5500	SYRINGE, HYPODERMIC, Luer Lock, Control, Glass, 3 ml	Type II
6515-00-827-3104	SYRINGE, HYPODERMIC, Luer Lock, Glass, 3 ml	Type I

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity as indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

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Navy - MS
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Preparing activity:
DLA-DM

Civil Agency Coordinating Activities:
GSA-FSS
VA-DMS

Project No. 6515-2052

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 each.

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DOCUMENT IDENTIFIER AND TITLE
GG-S-935A SYRINGES, HYPODERMIC, NEEDLE LOCK, GLASS, 3 ml (cc)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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